System Requirements Specification

1. Introduction

Purpose: provide homeowners and residents with an integrated residential security platform using motion, door/window and water-leak sensors, IP cameras, alarms, smart-device controllers, local storage and optional cloud services. Critical local detection and alarm functions remain available during Internet loss.

Scope: monitor security/safety events; arm/disarm zones; stream and record video; activate alarms; notify users; authenticate and authorize users; retain event history; control compatible locks/lights. Police, fire, medical and professional monitoring services are out of scope.

2. Overview

Context: a Central Control Unit is the local processing node. Zigbee sensors/controllers and IP/PoE cameras send events locally. Mobile, web and wall-panel users access the system through authenticated interfaces. Cloud services provide remote access, notifications, synchronization and backup.

Functions: SF-01 motion detection; SF-02 door/window monitoring; SF-03 water-leak detection; SF-04 video; SF-05 local alarm; SF-06 alerts; SF-07 arm/disarm; SF-08 authentication/RBAC; SF-09 remote access; SF-10 event storage; SF-11 smart-device control; SF-12 local operation during Internet loss.

Users: Homeowner/Admin (full configuration); Resident (monitoring, permitted cameras, arm/disarm); Maintenance Technician (installation, pairing, diagnosis and replacement with restricted private-data access).

3. Functional Requirements

FR-MOT-01 detect events from registered motion sensors. FR-MOT-02 create a security event for motion in an armed zone. FR-DW-01 detect door/window open/close. FR-WL-01 detect water leaks; FR-WL-02 notify authorized users. FR-CAM-01 support live IP-camera video; FR-CAM-02 allow authorized remote viewing; FR-CAM-03 support event-triggered recording; FR-CAM-05 indicate offline cameras. FR-ALM-01 activate local alarm for qualifying intrusion; FR-ALM-03 notify users. FR-CCU-01 process registered device events; FR-CCU-03 retain essential local detection, logging and alarm without cloud. FR-AUTH-01 require authentication; FR-AUTH-02 reject invalid credentials; FR-AUTH-03 provide Admin and Resident roles. FR-CTL-01 support compatible locks, lights and siren controllers.

4. Usability Requirements

UR-01 resident shall arm/disarm in no more than three UI actions. UR-02 UI shall display Disarmed, Armed Stay, Armed Away, Alert or Alarm. UR-03 alerts identify type, device/location and time. UR-04 at least 90% of representative first-time users shall arm/disarm without operational assistance. UR-06 recoverable errors shall provide a user-relevant recovery instruction.

5. Performance Requirements

PR-01 process normal sensor event and update local state within 1 second. PR-02 activate configured local alarm within 2 seconds after qualifying intrusion confirmation. PR-03 submit remote notification within 3 seconds of alarm activation when Internet is available. PR-04 support at least 50 registered sensors/controllers. PR-05 support at least 8 IP cameras subject to bandwidth/storage. PR-06 support at least 5 simultaneous authorized user devices. PR-07 support continuous 24-hour local operation. PR-08 restore essential operation within 120 seconds after a normal software restart.

6. System Interfaces

IF-01 low-power sensors use supported Zigbee. IF-02 cameras use local IP over Ethernet/PoE or approved Wi-Fi. IF-03 remote users use HTTPS. IF-04 cloud messaging uses HTTPS or MQTT over TLS. IF-05 loss of cloud/remote interface shall not disable essential local intrusion detection and alarm processing. Local database/storage uses a local service/API; critical equipment uses AC mains with UPS/battery where appropriate.

7. System Operations

HSI-01 critical arm/disarm, alarm acknowledgement and alarm status are visible. HSI-02 high-impact admin actions require confirmation. HSI-03 maintenance access is role-restricted. HSI-04 state is conveyed by text as well as color. MR-01 a failed supported sensor is replaceable without replacing the hub. MR-02 device health is visible. MR-04 trained technician replaces/re-registers a supported sensor within 30 minutes. REL-01 local intrusion/alarm works without Internet. REL-03 central unit target availability is at least 99.5% monthly excluding planned maintenance. REL-04 critical events during cloud outage are retained locally.

8. Modes / Physical / Adaptability / Environment

Modes: Disarmed, Armed Stay, Armed Away, Alert, Alarm, Maintenance, Offline/Degraded. PHY-01 hub is suitable for indoor residential installation; PHY-02 outdoor cameras use suitable enclosures; PHY-03 sensor batteries are accessible. AD-01 compatible sensors can be added without redesign; AD-02 architecture supports future growth to at least 100 devices subject to capacity. ENV-01 indoor electronics target 0 C to 40 C; ENV-03 battery sensors continue during AC interruption; ENV-05 retry/mesh/reconnection handles normal radio interference.

8.4 System Security

SEC-01 sensitive remote traffic is encrypted in transit. SEC-02 passwords are not stored in plaintext. SEC-03 RBAC separates Admin and Resident. SEC-04 admin configuration is restricted. SEC-05 auth failures are logged. SEC-06 significant actions are audited. SEC-07 secure updates reject unauthorized packages where supported. SEC-08 sensitive local data uses OS/application access controls and encryption for secrets/tokens. SEC-09 camera/event access follows role and household policy.

9. Verification

Verification methods: inspection, analysis, demonstration, functional integration testing, performance/load testing, security testing and fault injection. The Verification & Validation workbook contains objectives, priorities, requirement type, preconditions, numbered steps, an Expected outcome paired with every step, final expected result and requirement traceability.

Required coverage: motion sensor integration, door/window integration, water leak integration, camera/live view, valid and invalid login; user validation for motion alerts, door/window activation, remote live view, arm/disarm; load validation for 3+ simultaneous sensor triggers and 3+ concurrent user devices; offline-reliability scenarios are also included.

10. Assumptions and Dependencies

A-01 devices/network are installed correctly. A-02 remote access/push depend on Internet/cloud availability. A-03 essential local intrusion/alarm does not depend on Internet. A-04 users have supported devices/software. A-05 household maintains batteries. A-06 LAN bandwidth/storage are sufficient for configured video. A-07 cloud outage may remove remote functions but not local security. A-08 installation locations provide adequate signal, network, power and environmental protection.

Traceability Summary

Motion: FR-MOT -> VER-001, VAL-001. Door/window: FR-DW -> VER-002, VAL-002. Water leak: FR-WL -> VER-003. Camera/remote view: FR-CAM -> VER-004, VAL-003. Authentication/security: FR-AUTH/SEC -> VER-005, VER-006. Offline reliability: FR-CCU-03/REL -> VER-007, VAL-008. Arm/disarm usability: UR -> VAL-004, VAL-009. Load/concurrency: PR-01/PR-06 -> VAL-005, VAL-006.

Design basis: local-first hybrid edge + cloud. Weighted tradeoff matrices justify Zigbee for low-power sensors/controllers, PoE IP cameras for fixed video, and hybrid local+cloud server/manager architecture. Zigbee is retained as the architecture choice even where Z-Wave has a marginal raw score advantage because one shared mesh/coordinator reduces gateway complexity, integration effort, maintenance and multi-radio interoperability risk.